

Flood Impact Assessment and Dynamic Mapping

Factor	Dataset Needed	Where to Request in Sri Lanka	Format	Resolution	Purpose
Historical Flood Records	Past flood event records	Disaster Management Centre (dmc.gov.lk); DesInventar Sri Lanka (desinventar.net, country: LKA)	Tabular	Event-level	Historical severity benchmarking and model training data
River Water Level Data	River gauge water level readings	Irrigation Department of Sri Lanka (irrigation.gov.lk); Department of Hydrology; River Gauge Stations	Time-series (CSV/API)	Hourly / daily, per gauge station	Real-time flood threshold detection and rising-water alerts
Rainfall Data	Rain gauge network + historical rainfall records	Department of Meteorology (meteo.gov.lk)	Time-series (CSV/API)	Hourly / daily, per station	Primary trigger input for flood forecasting model
Digital Elevation Model (DEM)	Elevation / terrain model	Survey Department of Sri Lanka (survey.gov.lk); SRTM or ALOS PALSAR if unavailable locally	Raster (GeoTIFF)	<10m (requested); 30m (SRTM) / 12.5m (ALOS) fallback	Flood extent modeling, flow accumulation, and low-lying area identification
Land Use / Land Cover Data	Land use classification	Survey Department of Sri Lanka (survey.gov.lk); Ministry of Environment (env.gov.lk)	Raster / Vector	10m or 1:50,000 base	Surface runoff estimation; impervious surface and vulnerable land use identification
Satellite Imagery (Flood Extent)	Flood extent detection imagery	Copernicus Sentinel-1 & Sentinel-2 (ESA – browser.dataspace.copernicus.eu); USGS Landsat (earthexplorer.usgs.gov)	Raster (SAR / optical)	10m (Sentinel) / 30m (Landsat)	Actual flood extent mapping and validation of model predictions
Administrative Boundaries (Province, District, DS, GN)	Administrative boundary layers	Survey Department of Sri Lanka (survey.gov.lk); Department of Census & Statistics (statistics.gov.lk)	Vector (polygon)	Province / District / DS / GN division level	Spatial aggregation unit for combining all component datasets and reporting
River Network / Drainage Data	River and drainage network layers	Survey Department of Sri Lanka (survey.gov.lk); Irrigation Department (irrigation.gov.lk)	Vector (line)	1:50,000 topo base	Hydrological network modeling and downstream flood propagation
Historical Damage & Loss Records	Damage and loss records from past floods	Disaster Management Centre (dmc.gov.lk); DesInventar Sri Lanka (desinventar.net, country: LKA)	Tabular	Event / district level	Loss estimation, impact severity modeling, and damage benchmarking